Hierarchical Shell Manifold Theory: Operator Reconstruction, Spectral Identifiability, and No-Go Boundaries

Abstract

Hierarchical Shell Manifold Theory (HSMT) provides a graded operator architecture in which effective dynamics may be studied after projection onto a reduced hierarchy. In this work we isolate the inverse-spectral content of that architecture and separate reconstructible operator data from additional microscopic or geometric interpretations.

On a cyclic reduced sector, we represent the hierarchy by a half-line Jacobi operator

$$(J\psi)_N = \lambda_{N-1}\psi_{N-1} + b_N\psi_N + \lambda_N\psi_{N+1},$$

with real b_N and positive off-diagonal coefficients up to any finite termination depth. With a normalized cyclic contact vector, the boundary resolvent

$$m_0(z) = \langle e_0, (z - J)^{-1} e_0 \rangle$$

determines the cyclic spectral measure and hence the associated Jacobi coefficients. Moment-only reconstruction additionally requires the usual determinacy conditions. We develop this reconstruction explicitly and test it through finite-rank, long-tail, noisy, finite-resolution, and mixed continuous–discrete spectral benchmarks. These tests show that a stable finite Jacobi prefix can be reconstructed from controlled boundary spectral data, while the infinite asymptotic tail remains non-identifiable from finite noisy observations of the class considered.

We also derive the exact equivalence, within the cyclic Jacobi sector, between centered spectral symmetry and a constant Jacobi diagonal, providing a direct finite-depth falsification test for diagonal detuning. Finitely generated ladder models are then treated as candidate classes rather than derived dynamics. Toeplitz, canonical-bosonic, and SU$(1,1)$-type examples yield distinct cross-level relations, but finite-prefix data do not in general identify a unique microscopic algebra; in particular, normalized SU$(1,1)$ models approach the canonical-bosonic sequence arbitrarily closely at any fixed finite depth as the representation parameter increases.

From a given Jacobi operator we further construct, conditionally on a strictly positive spectral-edge solution, an operator-induced radial reference geometry via a ground-state/Doob transform. No identification with physical space is made. Explicit counterexamples show that asymptotically constant hopping does not determine a unique reference volume-growth exponent.

The resulting analysis establishes a set of identifiability and selection boundaries. The present HSMT architecture does not determine a unique asymptotic Jacobi law, centered resonance, microscopic ladder algebra, physical spatial geometry, Lorentzian completion, or fundamental experimental target. We therefore distinguish throughout among reconstruction, selection, and physical interpretation. The principal positive result is a controlled inverse-spectral framework for finite effective hierarchy dynamics together with explicit criteria for candidate falsification and explicit no-go boundaries for stronger inferences.

Logical-status notation. Throughout the paper we use the following labels: [D] derived from the explicitly stated assumptions; [DC] derived conditionally once additional stated assumptions are imposed; [R] numerical or reconstruction result obtained in a declared benchmark; [C] candidate principle, interpretation, or model class not derived from the current architecture; [O] open problem or missing physical input; and [NG] no-go result showing that a claimed conclusion does not follow from the present assumptions. These labels track logical status rather than subjective confidence.

1 Introduction

Hierarchical Shell Manifold Theory was introduced as a graded operator framework intended to separate the organization of hierarchical sectors from the later emergence of effective physical structure. The foundational architecture is taken here as input. We do not modify that architecture in this paper. Instead, we ask a narrower question: once a reduced cyclic dynamical sector has been obtained, what information about that sector can be reconstructed from spectral response, and which stronger conclusions remain underdetermined?

This question requires a distinction that will be maintained throughout:

$$\boxed{\text{reconstruction} \neq \text{selection} \neq \text{interpretation.}}$$

Reconstruction asks whether operator coefficients can be inferred from observable spectral data. Selection asks whether the theory itself determines a particular law for those coefficients. Interpretation asks whether a mathematically reconstructed structure should be identified with physical space, spacetime, or another observable sector. These are logically separate problems.

The reduced dynamics considered below are represented by a half-line Jacobi operator. This representation is a reduction of a cyclic sector; it is not derived as the unique nearest-neighbor form of the full graded architecture. Throughout the Jacobi reduction the contact vector e_0 is normalized and the off-diagonal coefficients are fixed in the standard positive convention $\lambda_N > 0$ up to any finite termination depth. The boundary resolvent defines a scalar spectral observable from which the cyclic Jacobi coefficients are exactly recoverable under the conditions stated below. This provides a precise setting in which to study what finite spectral information can and cannot determine.

The first objective of this paper is constructive. We derive the inverse relation between the boundary resolvent, the spectral measure, and the Jacobi coefficients, and we validate finite-prefix reconstruction through controlled synthetic benchmarks. These benchmarks include nonconstant off-diagonal weights, diagonal detuning, finite spectral resolution, noise, long-range coefficient drift, explicit asymptotic-tail nonuniqueness, and spectral measures containing both continuous and discrete components.

The second objective is falsificatory. Several structures that might otherwise be mistaken for predictions of the graded architecture are tested against explicit counterfamilies. In particular, we examine centered resonance, finite-generativity ladder models, asymptotic Jacobi laws, and operator-induced reference geometry. Where the current assumptions admit inequivalent alternatives, we record that nonuniqueness as a no-go result rather than selecting a preferred continuation by simplicity or by agreement with a desired physical outcome.

The third objective is methodological. A conditional ground-state/Doob construction can associate a radial reference geometry with an admissible Jacobi operator. That construction is useful, but mathematical canonicity does not establish physical identification. Likewise, a scalar boundary

spectral measure determines a Jacobi/Krylov chain relative to its contact vector but does not by itself prove that the reconstructed chain coincides with an independently postulated physical HSMT grade basis. These distinction lines are central to the paper.

The paper consequently addresses three questions:

1. What finite hierarchy dynamics are reconstructible from boundary spectral data?
2. Which candidate microscopic or dynamical structures can be falsified using the reconstructed coefficients?
3. Which stronger microscopic, geometric, temporal, or experimental conclusions remain non-identifiable under the present architecture?

The answers are asymmetric. The finite inverse problem is well controlled: exact cyclic spectral data determine the associated Jacobi coefficients, and finite noisy data can recover a stable finite prefix under declared resolution and model conditions. The infinite continuation is not determined by finite data. Centered spectral symmetry yields a sharp test of diagonal detuning, but centered resonance itself is not selected by the current architecture. Finitely presented ladder algebras generate genuine cross-level predictions, but no existing HSMT principle selects one such algebra. A radial reference geometry can be constructed conditionally from the Jacobi operator, but neither its asymptotic exponent nor its identification with physical space follows from finite spectral reconstruction alone.

Several questions that previously appeared to require further calculation are therefore shown instead to require additional physical input. The absence of a derivation is treated as a substantive result when explicit counterfamilies demonstrate that the present assumptions admit inequivalent continuations.

This paper does not derive a physical spatial dimension, Lorentzian signature, temporal arrow, cosmological redshift mechanism, CMB shell boundary, dark-sector interpretation, or a value of α_{H}. Historical derivational chains not re-established under the present status rules are likewise excluded. The aim is narrower: to state precisely what the current operator architecture does determine, what can be reconstructed from spectral data, and where its present identifiability boundaries lie.

2 Reduced Jacobi dynamics

Let

$$\mathcal{H}_{\mathrm{cyc}} = \overline{\mathrm{span}}\{e_0, Je_0, J^2e_0, \ldots\}$$

be the cyclic subspace generated by a normalized contact vector e_0. The recurrence index is denoted N for economy. In a boundary spectral reconstruction it labels the Jacobi/Krylov chain associated with the chosen contact vector; identifying it with an independently defined HSMT grade requires a separate physical bridge.

Assuming either that e_0 lies in the domain of all powers of J required for the Lanczos/Krylov construction, or that the recurrence is constructed from the spectral measure, orthonormalization of the Krylov sequence gives a tridiagonal representation

$$(J\psi)_N = \lambda_{N-1}\psi_{N-1} + b_N\psi_N + \lambda_N\psi_{N+1}, \qquad N \geq 0, \tag{1}$$

with boundary convention $\lambda_{-1} = 0$. We take $b_N \in \mathbb{R}$. In an infinite irreducible half-line chain, $\lambda_N > 0$ for every N. In a finite-dimensional cyclic sector the recurrence terminates when some $\lambda_K = 0$; the positive convention applies only before that termination depth.

Proposition 2.1 (Cyclic Jacobi representation [DC]). *Let J be self-adjoint with normalized cyclic vector e_0. Under the domain or spectral-measure assumptions above, the restriction of J to its cyclic subspace is represented by a Jacobi operator with real diagonal coefficients and positive off-diagonal coefficients up to finite termination.*

The index in equation (1) initially labels Krylov levels. The representation alone does not establish that N is a physical spatial coordinate, an effective dimension, a spacetime coordinate, or an experimentally resolved HSMT grade. If an independently specified HSMT reduction identifies the cyclic basis with a hierarchy basis, then the same index can also label that reduced hierarchy. Boundary spectroscopy by itself does not supply this identification.

2.1 Diagonal and off-diagonal data

The sequence $\{b_N\}$ contains diagonal energies or detunings in the reduced representation. The sequence $\{\lambda_N\}$ contains the nearest-neighbor matrix elements generated by tridiagonalization. No asymptotic law is assumed for either sequence. In particular, we do not impose constant hopping, square-root growth, linear growth, zero diagonal, or constant diagonal unless such forms are introduced later as explicit candidate models.

This is the reconstruction/selection firewall: the inverse problem reconstructs coefficients; it does not select their law.

2.2 Relation to HSMT projection

The foundational HSMT architecture contains projected structures of the schematic form PHP, PHQ, QHP, and QHQ. A one-channel Feshbach reduction naturally produces a scalar self-energy containing a matrix element of a Q-sector resolvent [3]. This motivates the boundary-resolvent problem studied here. The existence of such a projected resolvent does not itself imply that the Q-sector is already represented by a unique physical Jacobi chain. Rather, one specifies a contact vector, generates its cyclic subspace, and tridiagonalizes that subspace.

The Jacobi operator used here is therefore a controlled reduced representation, not a new frozen postulate about the full HSMT architecture. Standard Jacobi and orthogonal-polynomial background may be found in Refs. [1, 2].

2.3 Self-adjointness scope

For bounded coefficients,

$$\sup_N |b_N| < \infty, \qquad \sup_N \lambda_N < \infty,$$

the Jacobi matrix defines a bounded self-adjoint operator on $\ell^2(\mathbb{N}_0)$. More general unbounded Jacobi operators require domain control. The inverse-spectral results below are stated for self-adjoint realizations for which e_0 is cyclic. We do not identify boundedness with physical necessity; earlier bounded-hop conditions are sufficient analytic constructions, not universal selection principles.

2.4 Contact spectral measure

By the spectral theorem, the normalized cyclic vector e_0 defines a probability measure

$$d\mu(E) = d\langle e_0, P_J(E)e_0\rangle, \qquad \int_{\mathbb{R}} d\mu(E) = 1.$$

For suitable functions f,

$$\langle e_0, f(J)e_0\rangle = \int_{\mathbb{R}} f(E)\, d\mu(E).$$

Whenever finite, its moments are

$$\mu_n = \langle e_0, J^n e_0\rangle = \int_{\mathbb{R}} E^n\, d\mu(E).$$

The exact measure may contain absolutely continuous, discrete, singular-continuous, or mixed components. The numerical benchmarks below test absolutely continuous densities and discrete atoms; they do not validate generic singular-continuous reconstruction.

3 Boundary resolvent and exact inverse problem

For $z \notin \sigma(J)$ define

$$m_0(z) = \langle e_0, (z - J)^{-1} e_0\rangle. \tag{2}$$

With this convention,

$$\operatorname{Im} z > 0 \quad \Longrightarrow \quad \operatorname{Im} m_0(z) < 0.$$

The normalization of e_0 gives the large-z expansion

$$m_0(z) = \frac{1}{z} + \frac{\mu_1}{z^2} + \frac{\mu_2}{z^3} + O(z^{-4}), \qquad z\, m_0(z) \to 1.$$

The function is the Stieltjes transform

$$m_0(z) = \int_{\mathbb{R}} \frac{d\mu(E)}{z - E}. \tag{3}$$

Exact knowledge of m_0 determines $d\mu$ through Stieltjes inversion together with discrete residue data.

3.1 Continued fraction and stripping

For the Jacobi operator,

$$m_0(z) = \cfrac{1}{z - b_0 - \cfrac{\lambda_0^2}{z - b_1 - \cfrac{\lambda_1^2}{z - b_2 - \ddots}}}. \tag{4}$$

Writing m_N for the stripped boundary resolvent at depth N,

$$m_N(z) = \frac{1}{z - b_N - \lambda_N^2 m_{N+1}(z)},$$

so that

$$m_{N+1}(z) = \frac{z - b_N - m_N(z)^{-1}}{\lambda_N^2}. \tag{5}$$

Proposition 3.1 (Exact boundary-spectral reconstruction [D])**.** *Let J be a self-adjoint half-line Jacobi operator with normalized cyclic boundary vector e_0 and positive recurrence convention. Exact knowledge of the boundary spectral measure, equivalently the exact boundary m-function, determines the Jacobi recurrence coefficients in the cyclic sector. Moment determinacy is required only if the measure itself is to be inferred from moments alone.*

3.2 First coefficients from moments

The first moments satisfy

$$\mu_0 = 1, \qquad \mu_1 = b_0, \qquad \mu_2 = b_0^2 + \lambda_0^2,$$

and hence

$$b_0 = \mu_1, \qquad \lambda_0^2 = \mu_2 - \mu_1^2. \tag{6}$$

The third moment is

$$\mu_3 = b_0^3 + 2b_0\lambda_0^2 + b_1\lambda_0^2. \tag{7}$$

3.3 Measure knowledge versus moment knowledge

Two inverse questions must be separated. If the exact m-function or exact spectral measure is given, then the chain

$$m_0 \longrightarrow d\mu \longrightarrow \{b_N, \lambda_N\}$$

is fixed on the cyclic Jacobi sector. If only a moment sequence $\{\mu_n\}$ is given, then a moment-determinacy condition is additionally required to infer a unique measure from those moments. Moment indeterminacy is therefore distinct from nonuniqueness of the Jacobi coefficients once an exact spectral measure has already been specified.

3.4 Centered-symmetry preview

If the exact contact measure is symmetric about β, then the shifted measure in $E - \beta$ is even. For the irreducible Jacobi recurrence this implies $b_N = \beta$ throughout the supported recurrence chain. The full equivalence, together with its chiral form, is proved in section 5. No resonance principle is assumed here.

3.5 What exact inversion does not imply

Exact inverse reconstruction does not imply that the coefficients obey a simple asymptotic law, that the Jacobi index is a physical HSMT grade, that the reconstructed operator is the unique microscopic Hamiltonian, that the spectral measure is purely absolutely continuous, that finite noisy data determine infinitely many coefficients, or that the resulting Jacobi operator determines physical geometry.

4 Application A1: finite spectral reconstruction and identifiability

The exact inverse problem assumes complete knowledge of the boundary spectral measure or m-function. Any physical measurement instead provides finite-resolution, finite-bandwidth, and noisy data. Application A1 therefore tests which parts of the exact inverse structure remain practically recoverable under controlled synthetic conditions. The purpose is not to select a microscopic HSMT law. It is to validate reconstruction machinery and expose identifiability limits.

4.1 Finite-rank benchmark family

We first use the declared finite-rank family

$$\lambda_N = \begin{cases} \lambda_0, & N = 0, \\ \lambda_1, & N = 1, \\ \lambda_\infty, & N \geq 2, \end{cases} \qquad b_N = \begin{cases} b_0, & N = 0, \\ b_1, & N = 1, \\ 0, & N \geq 2. \end{cases}$$

The free tail has

$$m_\infty(z) = \frac{z - \sqrt{z^2 - 4\lambda_\infty^2}}{2\lambda_\infty^2}, \tag{8}$$

with the branch fixed by $m_\infty(z) \sim z^{-1}$. Backward continued fractions then give

$$m_1(z) = \frac{1}{z - b_1 - \lambda_1^2 m_\infty(z)}, \qquad m_0(z) = \frac{1}{z - b_0 - \lambda_0^2 m_1(z)}.$$

4.2 A1-S: symmetric nonconstant benchmark

Declared parameters are

$$\lambda_\infty = 1, \qquad \lambda_0 = 0.8, \qquad \lambda_1 = 1.15, \qquad b_0 = b_1 = 0.$$

The tail is free while the boundary couplings are nonconstant. Noiseless parametric inversion recovers the declared finite-rank parameters to numerical precision. The benchmark therefore demonstrates that exact spectral symmetry does not imply constant λ_N. No off-support discrete atom was detected in the declared search interval for this family. These are benchmark results, not HSMT predictions.

4.3 A1-D: diagonal detuning

Use the same off-diagonal family with

$$b_0 = 0.2, \qquad b_1 = -0.1.$$

Noiseless parametric recovery reconstructs the declared coefficients to numerical precision. The contact measure is strongly asymmetric. Here

$$b_1 - b_0 = -0.3, \qquad \lambda_0^2 = 0.64,$$

so the centered third-moment identity derived later gives

$$\nu_3 = \lambda_0^2(b_1 - b_0) = -0.192.$$

The benchmark moments reproduce this value. Thus A1-D provides a declared finite-rank consistency check for finite-depth diagonal detuning.

4.4 Finite resolution and noise

Finite-resolution data access $m(E+i\eta)$ rather than a strict boundary value. The principal benchmark resolution is

$$\eta = 0.02.$$

Finite η broadens spectral features and regularizes inversion. Synthetic complex response noise was applied at declared fractional scales 10^{-4}, 10^{-3}, and 10^{-2}. In the finite-rank family, parameter recovery remains stable across the tested range, with errors tracking the imposed noise scale. This is [R] for the declared low-dimensional parametric family, not a theorem of universal inverse stability.

4.5 A1-LR: long-range drift

The long-tail truth is declared as

$$\lambda_N = 1 - \frac{0.35}{(N+2)^{1.3}}, \qquad b_N = \frac{0.15}{(N+2)^{1.1}}. \tag{9}$$

The estimator is not given the exponents or amplitudes. It fits a finite Jacobi prefix with a free continuation model. In the frozen benchmark record, a depth-$K = 6$ fit produced held-out relative m-function error approximately

$$3.46 \times 10^{-3},$$

while $K = 10$ reduced the error to approximately

$$6.94 \times 10^{-4}.$$

Increasing depth reduced finite-truncation bias in this benchmark, while coefficients nearest the terminator remained more sensitive to continuation mismatch.

4.6 Stable reconstruction depth

Finite data do not justify reporting every fitted coefficient. Let K_{stable} denote the maximum contiguous coefficient depth satisfying a declared stability criterion under noise perturbation, model-depth variation, and held-out response validation. The numerical threshold is a benchmark convention rather than a theorem. In the frozen A1 long-tail tests at approximately one-percent response noise, a stable contiguous prefix persisted only to finite depth; deeper coefficients became dominated by terminator/model bias.

4.7 Explicit asymptotic-tail nonuniqueness

Tail A is the truth in equation (9). Tail C is defined to agree exactly with Tail A through indices $N = 0, \ldots, 9$ and then continue with

$$\lambda_N^C = 1 - \frac{0.0542852834}{(N+2)^{0.55}}, \qquad b_N^C = \frac{0.0555160759}{(N+2)^{0.70}}. \tag{10}$$

The two infinite operators therefore have the same first ten Jacobi layers but inequivalent asymptotic decay exponents. Their exact boundary m-functions differ only weakly at finite resolution. In the declared calculations, the relative differences were approximately

$$5.08 \times 10^{-4} \quad (\eta = 0.005),$$
$$3.89 \times 10^{-4} \quad (\eta = 0.02),$$

and

$$2.44 \times 10^{-4} \quad (\eta = 0.05).$$

These differences lie below the tested 10^{-3} and 10^{-2} noise scales.

Proposition 4.1 (Finite-data/infinite-tail identification no-go [NG])**.** *Finite noisy boundary spectral data of the class considered here do not uniquely determine the asymptotic Jacobi tail. The Tail A/Tail C construction is an operational witness: the two infinite Jacobi operators agree through the reconstructed prefix yet possess inequivalent asymptotic laws and remain unresolved at the declared resolutions and noise levels.*

4.8 Terminator bias

A finite-prefix inverse model necessarily imposes some continuation or terminator. If that continuation is incorrect, deeper fitted coefficients can absorb the mismatch. In A1-LR the dominant error at sufficiently high signal quality can therefore be terminator bias rather than raw measurement noise. Operationally, only coefficients stable under terminator variation should be reported, and an asymptotic power law should never be inferred merely because the chosen terminator has one.

4.9 A1-P: mixed continuous–discrete benchmark

Use the declared parameters

$$\lambda_\infty = 1, \qquad \lambda_0 = 0.8, \qquad \lambda_1 = 1.15, \qquad b_0 = 2.5, \qquad b_1 = -0.1.$$

The essential free-tail band is $[-2, 2]$. The exact continued-fraction pole condition gives an upper off-band pole

$$E_p \approx 2.77656525146$$

with contact spectral weight

$$w_p \approx 0.86623881586.$$

The remaining continuous weight is approximately

$$w_{\mathrm{cont}} \approx 0.13376118400,$$

so that $w_p + w_{\mathrm{cont}} \approx 1$. The reported pole location and weight are exact outputs of the declared continued-fraction calculation for the stated finite-rank parameters. The majority of the contact spectral mass in this benchmark is discrete.

Proposition 4.2 (Continuous-density/full-measure identification no-go [NG])**.** *The absolutely continuous spectral density does not determine the full contact spectral measure when discrete or singular components are present.*

At finite η, an isolated atom contributes a Lorentzian-broadened feature with scale

$$\rho_\eta(E_p) \approx \frac{w_p}{\pi\eta}.$$

For $\eta = 0.02$, this gives approximately 13.79, consistent with the broadened benchmark. A narrow continuous resonance can nevertheless also produce a finite-width peak.

Proposition 4.3 (Finite-η peak/discrete-atom identification no-go [NG])**.** *A peak in a single finite-resolution spectral density is not by itself evidence for a true point mass. Exact analytic $m(z)$ can identify a real pole by residue, whereas a single finite-η profile cannot.*

Within the declared finite-rank family, noisy parametric pole reconstruction is stable. In 30 realizations, the frozen 95th-percentile errors were approximately

noise	$\lvert\Delta E_p\rvert$	$\lvert\Delta w_p\rvert$
10^{-4}	1.48×10^{-6}	6.92×10^{-6}
10^{-3}	1.44×10^{-5}	4.63×10^{-5}
10^{-2}	1.46×10^{-4}	7.33×10^{-4}

for the declared parametric protocol. This does not establish equivalent stability for arbitrary nonparametric mixed measures.

4.10 A1 status

Application A1 supports four distinct conclusions. First, stable finite Jacobi prefixes can be reconstructed in the declared benchmark families. Second, finite diagonal detuning and discrete pole data are detectable in appropriately parameterized settings. Third, continuous density is not the full spectral measure and a finite-η peak is not proof of a discrete atom. Fourth, finite spectral data of the tested class do not determine the infinite Jacobi continuation or its downstream asymptotic reference geometry.

A1 does not show that any synthetic coefficient law describes nature, that any benchmark tail is selected by HSMT, that a reconstructed Krylov chain is a physical HSMT grade chain, or that finite coefficients determine physical dimension.

5 Centered resonance, spectral symmetry, and detuning

A statement such as $b_N = 0$ depends on the choice of energy origin. The invariant condition is whether there exists a constant β such that

$$b_N = \beta$$

for every supported recurrence level. We call this *centered resonance*. After the shift $J \mapsto J - \beta I$, a centered-resonant Jacobi operator has zero diagonal.

5.1 Parity and chiral form

Define

$$\Gamma e_N = (-1)^N e_N.$$

If $J = B + A + A^*$, with $Be_N = b_N e_N$ and A the forward hopping operator, then

$$\Gamma B \Gamma = B, \qquad \Gamma(A + A^*)\Gamma = -(A + A^*).$$

Hence

$$\Gamma(J - \beta I)\Gamma = -(J - \beta I)$$

if and only if $b_N = \beta$ at every supported level.

Proposition 5.1 (Centered diagonal/chiral equivalence [D])**.** *For an irreducible Jacobi chain with parity $\Gamma e_N = (-1)^N e_N$, constant diagonal $b_N = \beta$ is equivalent to centered chiral anticommutation $\Gamma(J - \beta I)\Gamma = -(J - \beta I)$, up to the termination depth in a finite cyclic chain.*

5.2 Exact spectral symmetry

If $b_N = \beta$, then $J - \beta I$ anticommutes with Γ. Since $\Gamma e_0 = e_0$, the contact spectral measure satisfies

$$d\mu(\beta + x) = d\mu(\beta - x),$$

and the boundary resolvent obeys

$$m_0(\beta + z) = -m_0(\beta - z) \tag{11}$$

where both sides are defined.

Conversely, suppose the exact normalized contact measure is symmetric about β. The shifted measure is even, so the corresponding orthonormal polynomials alternate parity. Since

$$b_N = \int_{\mathbb{R}} E\, p_N(E)^2\, d\mu(E),$$

we have

$$b_N - \beta = \int_{\mathbb{R}} (E - \beta) p_N(E)^2\, d\mu(E) = 0.$$

No moment-determinacy assumption is required because the exact measure itself is given.

Proposition 5.2 (Centered spectral symmetry/constant-diagonal equivalence [D])**.** *For a normalized cyclic Jacobi measure with the standard positive recurrence convention, exact symmetry about β is equivalent to $b_N = \beta$ throughout the supported recurrence chain. Combined with the preceding proposition, centered spectral symmetry, constant Jacobi diagonal, and centered chiral anticommutation are equivalent within the cyclic Jacobi representation.*

Corollary 5.3 (Resonance center [D])**.** *For an exactly centered-symmetric normalized contact measure,*

$$\beta = \mu_1 = b_0.$$

5.3 Gauge-invariant detuning

Under $J \mapsto J + cI$, all b_N shift by c. We therefore define

$$\delta_N = b_N - b_0, \qquad \Delta_N = b_{N+1} - b_N.$$

Centered resonance is equivalent to $\delta_N = 0$ at every level.

Let ν_n denote centered moments about $\beta_0 = b_0 = \mu_1$. Using equations (6) and (7),

$$\nu_2 = \lambda_0^2, \qquad \nu_3 = \lambda_0^2(b_1 - b_0).$$

Hence, when $\nu_2 > 0$,

$$b_1 - b_0 = \frac{\nu_3}{\nu_2}. \tag{12}$$

Corollary 5.4 (First detuning from spectral skewness [D])**.** *The first energy-shift-invariant diagonal detuning is determined directly by the centered third spectral moment through equation* (12)*.*

For A1-D, $b_1 - b_0 = -0.3$ and $\lambda_0^2 = 0.64$, giving $\nu_3 = -0.192$ in agreement with the synthetic benchmark.

5.4 Centered Resonant Immanence as a candidate

[C] **Centered Resonant Immanence** is the candidate that there exists a common β such that $b_N = \beta$ throughout the physical reduced hierarchy. The equivalences above make this candidate spectrally testable. They do not derive it from HSMT.

The explicitly frozen forms of grading, bipartite off-diagonal incidence, self-adjointness, ordinary complex-conjugation invariance of a real Jacobi matrix, correspondence/Fock organization, intergrade immanence, Gaussian grade locality, and Feshbach resolvent structure do not force a constant diagonal. A bounded counterfamily is

$$J_b = A_\lambda + A_\lambda^* + \sum_N b_N P_N$$

with positive admissible λ_N and bounded nonconstant real b_N. It preserves the stated structures while violating centered chiral symmetry.

Proposition 5.5 (Resonance-selection no-go [NG])**.** *The explicitly frozen forms of the structures listed above do not force $b_N = \beta$. Centered Resonant Immanence therefore remains a candidate rather than a derived consequence of the current HSMT architecture.*

Graph bipartiteness alone constrains off-diagonal incidence and does not forbid onsite terms. Likewise the Feshbach effective operator

$$H_{\text{eff}}(z) = PHP + PHQ(z - QHQ)^{-1}QHP$$

can generate energy-dependent diagonal self-energy corrections unless an additional exact symmetry constrains them.

5.5 Finite-depth falsification and global limits

If a stable finite reconstruction resolves some $b_N - b_0 \neq 0$, then centered resonance is falsified over the observed cyclic depth. Conversely, agreement $b_N \approx b_0$ for $N \leq K$ cannot establish equality for all later N.

Proposition 5.6 (Finite-depth/global-resonance identification no-go [NG])**.** *Finite reconstruction of a centered diagonal over a finite Jacobi prefix does not establish global centered resonance of an infinite cyclic chain, because inequivalent continuations may begin beyond the observed prefix.*

Even exact centered spectral symmetry of one cyclic sector establishes only the corresponding sector symmetry. Other orthogonal sectors of the full Hamiltonian may break it. A scalar boundary spectral response therefore does not by itself establish a global chiral symmetry of the full microscopic theory.

A direct response diagnostic is

$$R_\beta(z) = m_0(\beta + z) + m_0(\beta - z),$$

which vanishes identically under exact centered symmetry. A covariance-aware finite-data version can test this condition without imposing any particular off-diagonal ladder law.

6 Finite-generativity candidate classes

Inverse reconstruction determines coefficients from spectral data but does not determine a microscopic law generating them. A stronger microscopic proposal should therefore make cross-level predictions that can be tested against the reconstructed sequence.

[C] **C22 finite ladder-algebra search class.** A model belongs to C22 if the hierarchy is generated from finite algebraic or representation data; successor and predecessor operators satisfy finitely presented relations; the reduced Hamiltonian is constructed from a finite set of algebra generators and parameters; Jacobi matrix elements follow from representation-theoretic matrix elements; no arbitrary function of a number operator is inserted merely to reproduce a desired coefficient sequence; and the model predicts falsifiable cross-level relations among b_N and λ_N. C22 is a candidate search class, not a derived HSMT principle.

Remark 6.1 (Finite symbols are not finite microscopic information [NG])**.** *Writing $H = f(\hat{N})$ or $\lambda_N = F(N)$ does not constitute finite microscopic compression if the function itself contains an arbitrary infinite sequence. Compact notation does not imply finite microscopic generativity.*

6.1 Toeplitz/isometric example

If the C15 correspondence candidate is adopted, one has a Fock-type hierarchy

$$\mathcal{F}(X) = \bigoplus_{N \geq 0} X^{\otimes N}.$$

Along a normalized cyclic tensor ray, canonical creation satisfies $T_\xi |N\rangle = |N+1\rangle$. The conditional Hamiltonian

$$H_T = \beta I + g(T_\xi + T_\xi^*)$$

therefore gives

$$b_N = \beta, \qquad \lambda_N = g.$$

This is [DC] if the Hamiltonian is adopted. C15 alone does not identify it as the physical Hamiltonian.

6.2 Bosonic CCR example

For normalized number states,

$$[a, a^\dagger] = I, \qquad a^\dagger|N\rangle = \sqrt{N+1}|N+1\rangle, \qquad a|N\rangle = \sqrt{N}|N-1\rangle.$$

The Hamiltonian

$$H_{\mathrm{CCR}} = \beta I + \omega a^\dagger a + g(a + a^\dagger)$$

gives

$$b_N = \beta + \omega N, \qquad \lambda_N = g\sqrt{N+1}.$$

This is a finite generative law conditional on adopting the CCR algebra. No bosonic statistics or CCR principle is derived by HSMT.

6.3 SU$(1,1)$ example

Take

$$[K_0, K_\pm] = \pm K_\pm, \qquad [K_+, K_-] = -2K_0.$$

In the positive discrete series with $k > 0$,

$$K_0|N\rangle = (N+k)|N\rangle,$$

$$K_+|N\rangle = \sqrt{(N+1)(N+2k)}|N+1\rangle,$$
$$K_-|N\rangle = \sqrt{N(N+2k-1)}|N-1\rangle.$$

Then

$$H_{\mathrm{SU}} = \beta I + \alpha K_0 + g(K_+ + K_-)$$

gives

$$b_N = \beta + \alpha(N+k), \qquad \lambda_N = g\sqrt{(N+1)(N+2k)}.$$

Again, this is [DC] once the algebraic model is adopted. No current HSMT principle selects the algebra or k.

Proposition 6.2 (Finite generativity does not select a unique Jacobi law [NG])**.** *Finite microscopic generativity alone does not select a unique Jacobi law. Toeplitz/isometric, CCR, and* SU$(1,1)$ *ladder systems are all finitely presented but yield mutually inequivalent cross-level matrix elements.*

The same graded Hilbert space can support distinct successor normalizations, for example

$$S|N\rangle = |N+1\rangle, \quad a^\dagger|N\rangle = \sqrt{N+1}|N+1\rangle, \quad K_+|N\rangle = \sqrt{(N+1)(N+2k)}|N+1\rangle.$$

Thus hierarchy alone does not select the ladder normalization.

Proposition 6.3 (Microscopic ladder-algebra selection no-go [NG]). *The explicitly frozen HSMT structures do not select among the Toeplitz/isometric, CCR,* $\mathrm{SU}(1,1)$*, or more general finite-generative C22 ladder laws. C22 therefore remains a falsifiable search class rather than a derived microscopic sector.*

Requiring bounded hopping also does not select constant Toeplitz weights. Bounded nonconstant examples include

$$\lambda_N = 1 + \frac{0.2}{N+2}$$

and

$$\lambda_N = 1 + 0.1(-1)^N.$$

Hence boundedness is an analytic condition, not a unique microscopic selector.

7 Application A2: finite-prefix model discrimination

Application A2 asks a different question from A1. A1 asks whether a finite Jacobi prefix can be reconstructed. A2 asks which finite-generativity candidates can be falsified once such a prefix is available. The intended output is candidate-specific rejection, not forced microscopic classification.

7.1 Off-diagonal finite-difference laws

Define

$$x_N = \lambda_N^2.$$

For the minimal candidate families,

$$\text{Toeplitz:} \qquad x_N = g^2, \qquad \Delta x_N = 0,$$

$$\text{CCR:} \qquad x_N = g^2(N+1), \qquad \Delta^2 x_N = 0,$$

$$\mathrm{SU}(1,1): \qquad x_N = g^2(N+1)(N+2k), \qquad \Delta^3 x_N = 0, \qquad \Delta^2 x_N = 2g^2 > 0.$$

These provide direct finite-prefix falsifiers.

7.2 Normalized CCR/$\mathrm{SU}(1,1)$ boundary

Normalize $x_0 = 1$. For $\mathrm{SU}(1,1)$ this gives $g^2 = 1/(2k)$ and therefore

$$x_N^{\mathrm{SU}} = (N+1)\left(1 + \frac{N}{2k}\right). \tag{13}$$

The normalized CCR sequence is $x_N^{\mathrm{CCR}} = N+1$, so

$$\frac{x_N^{\mathrm{SU}}}{x_N^{\mathrm{CCR}}} - 1 = \frac{N}{2k}. \tag{14}$$

Proposition 7.1 (CCR as a boundary of normalized $\mathrm{SU}(1,1)$ [D]). *For every fixed* N*,*

$$\lim_{k\to\infty} x_N^{\mathrm{SU}} = x_N^{\mathrm{CCR}}.$$

Proposition 7.2 (Finite-precision CCR/SU(1, 1) identification no-go [NG])**.** *For any finite observed depth $N = 0, \ldots, K$ and any $\varepsilon > 0$, choosing*

$$k > \frac{K}{2\varepsilon}$$

places the normalized SU(1, 1) *coefficient sequence within relative coefficient tolerance ε of the CCR sequence over the entire observed prefix. Finite noisy Jacobi data therefore do not uniformly separate CCR from the large-k* SU(1, 1) *family.*

For normalized SU(1, 1),

$$\Delta^2 x_N = \frac{1}{k}.$$

Hence unresolved curvature does not prove exact CCR; it means that positive SU(1, 1) curvature has not been resolved.

7.3 Hierarchical falsification tests

A natural sequence is:

1. test $\Delta x_N = 0$; rejection falsifies minimal Toeplitz hopping;
2. test $\Delta^2 x_N = 0$; rejection falsifies minimal CCR hopping;
3. test positive constant $\Delta^2 x_N$ together with $\Delta^3 x_N = 0$; rejection falsifies minimal SU(1, 1) hopping.

This is a hierarchy of candidate tests, not an automatic classifier.

The minimal diagonal laws are affine:

$$\text{Toeplitz: } b_N = \beta, \qquad \text{CCR: } b_N = \beta + \omega N, \qquad \text{SU}(1,1) : b_N = \beta + \alpha(N + k).$$

Thus all three satisfy $\Delta^2 b_N = 0$. Statistically significant diagonal curvature falsifies all three minimal Hamiltonians simultaneously, while leaving more general C22 models open.

7.4 Covariance-aware curvature test

Let the reconstructed vector $x = (x_0, \ldots, x_{K-1})^T$ have covariance Σ_x, and let D_2 be the second-difference matrix. Then

$$c = D_2 x, \qquad V = D_2 \Sigma_x D_2^T.$$

For normalized minimal SU(1, 1), $c = \theta \mathbf{1}$ with $\theta = 1/k \geq 0$. When V is nonsingular and well conditioned, the generalized least-squares estimator is

$$\widehat{\theta} = \frac{\mathbf{1}^T V^{-1} c}{\mathbf{1}^T V^{-1} \mathbf{1}}. \tag{15}$$

When V is singular or ill conditioned, the ordinary inverse is replaced by a Moore–Penrose pseudoinverse or an equivalent regularized generalized least-squares procedure. No statistical conclusion is attached to $\widehat{\theta}$ without propagated uncertainty.

7.5 Coefficient-level synthetic outcomes

Synthetic coefficient noise was applied directly to λ_N in the declared form

$$\lambda_N^{\mathrm{obs}} = \lambda_N(1+\epsilon_N), \qquad \epsilon_N \sim \mathcal{N}(0,\sigma_\lambda^2),$$

with representative σ_λ values 10^{-4}, 10^{-3}, and 10^{-2}. Candidate fitting was performed on $x_N = (\lambda_N^{\mathrm{obs}})^2$. The detailed selection frequencies depend on depth, coefficient-noise model, candidate set, and scoring choice, and are therefore benchmark statistics rather than physical probabilities.

The robust qualitative outcomes were: minimal Toeplitz is strongly separated from the curved examples in the declared tests; strongly curved $\mathrm{SU}(1,1)$ examples are readily separated from CCR; large-k $\mathrm{SU}(1,1)$ becomes increasingly difficult to distinguish from CCR at finite depth and nonzero uncertainty; and increasing stable depth improves discrimination when nonzero curvature is actually present. These behaviors are the expected operational manifestation of the finite-precision proposition above.

Higher finite differences amplify uncertainty, so covariance propagation should accompany candidate tests.

Proposition 7.3 (Finite-prefix/unique microscopic-algebra identification no-go [NG])**.** *Agreement of a finite reconstructed Jacobi prefix with one candidate ladder law does not establish that candidate as the unique underlying microscopic algebra. Finite data can falsify predicted cross-level relations, but do not generally prove microscopic uniqueness.*

Application A2 therefore validates a falsification framework, not a microscopic selection principle.

8 Operator-induced radial reference geometry

We now ask a distinct mathematical question: given a Jacobi operator, can one associate a reversible radial reference process without yet interpreting that process as physical space? The construction below is conditional. It requires a finite upper spectral edge and an explicitly assumed strictly positive spectral-edge solution. It is not available in this form for every Jacobi operator considered earlier.

8.1 Positive edge solution and ground-state transform

Let J be an irreducible half-line Jacobi operator with

$$E_* = \sup \sigma(J) < \infty.$$

Assume explicitly that there exists a strictly positive boundary-normalized solution

$$Jh = E_* h, \qquad h_N > 0. \tag{16}$$

The solution need not lie in ℓ^2; it may be a positive generalized edge solution.

For finitely supported f, define

$$(Q_{\mathrm{ref}} f)_N = \frac{1}{h_N}\big(J(hf)_N - E_* h_N f_N\big). \tag{17}$$

Using equation (16),

$$(Q_{\mathrm{ref}} f)_N = \lambda_{N-1} \frac{h_{N-1}}{h_N}(f_{N-1} - f_N) + \lambda_N \frac{h_{N+1}}{h_N}(f_{N+1} - f_N). \tag{18}$$

The off-diagonal rates are positive. Define

$$S_N^{\mathrm{ref}} = h_N^2, \qquad C_N^{\mathrm{ref}} = \lambda_N h_N h_{N+1}. \tag{19}$$

Then detailed balance holds across every edge.

Proposition 8.1 (C17a reversible reference construction [DC])**.** *Given an irreducible half-line Jacobi operator with finite upper spectral edge and an explicitly assumed strictly positive solution equation* (16)*, the transform equation* (17) *defines, on its natural test domain, a reversible nearest-neighbor birth–death operator with reference measure and conductances equation* (19)*. Additional analytic conditions may be required for a conservative Markov semigroup on the full infinite state space.*

The corresponding weighted Dirichlet form is

$$\mathcal{E}_{\mathrm{ref}}(f, f) = -\langle f, Q_{\mathrm{ref}} f \rangle_{S^{\mathrm{ref}}} = \sum_{N \geq 0} C_N^{\mathrm{ref}} |f_{N+1} - f_N|^2. \tag{20}$$

No factor of $1/2$ is missing because each undirected edge is counted once.

If $h_N \mapsto c h_N$, both S_N^{ref} and C_N^{ref} scale by c^2, while the generator rates remain unchanged. Thus the construction is canonical relative to J and the chosen edge solution up to overall reference-measure normalization whenever the boundary problem fixes the positive solution uniquely.

8.2 Reference volume growth

Define cumulative reference volume

$$V_{\mathrm{ref}}(R) = \sum_{N=0}^{R} h_N^2.$$

Assume

$$h_N \sim N^a L(N),$$

with L slowly varying and $a > -1/2$. Regular-variation asymptotics give

$$V_{\mathrm{ref}}(R) \sim \frac{R^{2a+1} L(R)^2}{2a+1}. \tag{21}$$

We define the reference volume-growth exponent

$$d_V^{\mathrm{ref}} = 2a + 1. \tag{22}$$

Proposition 8.2 (Reference volume exponent [DC])**.** *Under the stated regular-variation assumptions, the cumulative C17a reference measure has asymptotic growth exponent $d_V^{\mathrm{ref}} = 2a + 1$. This is a radial/reference exponent and is not identified with physical spatial dimension.*

If additionally $\lambda_N \to \lambda_\infty > 0$ and the coefficient variation is sufficiently regular for a continuum expansion with $r \simeq N$, then

$$Q_{\mathrm{ref}} \sim \lambda_\infty \left(\partial_r^2 + \frac{2a}{r} \partial_r \right). \tag{23}$$

This is a Bessel-type radial reference operator with parameter $D_{\mathrm{ref}} = 2a + 1$.

8.3 Free constant-weight example

For

$$\lambda_N = \lambda > 0, \qquad b_N = 0,$$

the upper edge is $E_* = 2\lambda$ and the positive normalized solution is

$$h_N = N + 1.$$

Thus

$$S_N^{\mathrm{ref}} = (N+1)^2, \qquad V_{\mathrm{ref}}(R) \sim \frac{R^3}{3}, \qquad D_{\mathrm{ref}} = 3,$$

and

$$C_N^{\mathrm{ref}} = \lambda(N+1)(N+2).$$

The continuum form is

$$Q_{\mathrm{ref}} \sim \lambda \left(\partial_r^2 + \frac{2}{r}\partial_r \right).$$

This is a Bessel-type radial reference process with reference parameter $D_{\mathrm{ref}} = 3$. It is not a derivation of physical three-space.

8.4 Asymptotically constant counterexample

Take

$$\lambda_N = 1 - \frac{1}{(N+1)(N+2)}, \qquad b_N = 0, \qquad h_N = (N+1)^2. \tag{24}$$

Direct substitution gives $Jh = 2h$, including at the boundary. Since $\lambda_N \to 1$, the essential upper edge is 2, and this example has $E_* = 2$. The reference measure is

$$S_N^{\mathrm{ref}} = (N+1)^4,$$

so

$$V_{\mathrm{ref}}(R) \sim \frac{R^5}{5}, \qquad D_{\mathrm{ref}} = 5.$$

Proposition 8.3 (Asymptotic-constant-weight/unique-reference-exponent no-go [NG])**.** *The condition $\lambda_N \to \lambda_\infty$ does not by itself determine a unique C17a reference volume exponent. The free constant-weight example gives $D_{\mathrm{ref}} = 3$, while equation* (24) *gives $D_{\mathrm{ref}} = 5$.*

8.5 Finite reconstruction and asymptotic C17a geometry

C17a asymptotic volume growth depends on the infinite spectral-edge solution and hence on the infinite Jacobi continuation. A finite stable prefix does not fix that continuation.

Proposition 8.4 (Finite-reconstruction/asymptotic-C17a no-go [NG])**.** *A finite reconstructed Jacobi prefix does not determine the asymptotic C17a reference volume exponent without independent information about the infinite Jacobi continuation.*

A specialized inverse-square-tail asymptotic relation is recorded in Appendix F; it is not required for the main reconstruction or identifiability results.

8.6 C17a versus C17b

[DC] **C17a** is the operator-induced reversible radial reference construction equation (19). [C] **C17b** is the proposed physical identification of that reference construction with physical grade volume or transport geometry. C17b does not follow from C17a. Mathematical canonicity does not establish physicality.

Even a radial operator of the form

$$\partial_r^2 + \frac{D_{\mathrm{ref}} - 1}{r}\partial_r$$

does not uniquely determine angular geometry, topology, tangential spectrum, radial/tangential scale, or the full metric. Different full geometric completions can possess the same radial operator. C17a therefore does not select full physical spatial geometry.

9 Identifiability and selection boundaries

The negative results collected here are not statements that the excluded structures are impossible. They state only that those structures do not follow from the assumptions currently present in HSMT.

9.1 Reconstruction does not select a microscopic law

Proposition 9.1 (Reconstruction/microscopic-law identification no-go [NG])**.** *Inverse reconstruction of Jacobi coefficients from spectral data does not by itself supply a forward microscopic law generating those coefficients. A forward law requires additional microscopic structure or independently supported empirical regularity beyond the inverse map.*

9.2 Asymptotic-tail selection

The explicitly frozen forms of grading, self-adjoint operator control, correspondence/Fock organization when C15 is adopted, inter-grade immanence, Gaussian grade locality, and Feshbach projection admit inequivalent asymptotic coefficient families.

Proposition 9.2 (Current-core asymptotic Jacobi-tail selection no-go [NG])**.** *The explicitly frozen HSMT architecture does not determine a unique asymptotic Jacobi-tail law. Reopening this question requires independent microscopic dynamics or independent asymptotic empirical information.*

Centered resonance is spectrally testable but not selected by the current architecture; see section 5. Likewise finite generativity does not select a unique ladder algebra; see section 7.

9.3 Reference geometry and physical geometry

Finite Jacobi reconstruction does not determine asymptotic C17a geometry by the proposition in section 8. More strongly, mathematical canonicity of C17a does not establish that the construction represents physical space.

Proposition 9.3 (Mathematical canonicity/physical-geometry identification no-go [NG])**.** *The canonical status of C17a relative to a given Jacobi operator does not establish that the resulting reference measure or conductances represent physical space. A physical identification requires independent operational or microscopic input.*

Likewise radial data do not uniquely determine full geometry.

Proposition 9.4 (Radial/full-geometry identification no-go [NG])**.** *A radial C17a operator does not uniquely determine a full spatial geometry. Additional angular/support structure and soldering data are required.*

9.4 Lorentzian and temporal completion

The same positive reference operator L may enter inequivalent temporal systems, for example

$$i\partial_t\psi = L\psi, \qquad (-\partial_t^2 + c^2 L)\phi = 0,$$

or a Euclidean/diffusive evolution. Self-adjointness and unitary evolution therefore do not by themselves select a Lorentzian principal symbol. Lorentzian signature, if independently supplied, also does not by itself select a temporal arrow.

Proposition 9.5 (Current-core Lorentzian/temporal selection no-go [NG])**.** *The current HSMT architecture does not determine a unique spacetime time coordinate, Lorentzian signature, causal cone, temporal/spatial calibration, or fundamental temporal arrow.*

9.5 Experimental-interface selection

The Feshbach formalism supplies a response once a physical decomposition P, Q and coupling QHP have been identified. The current architecture does not identify the laboratory probe, the hidden physical HSMT contact sector, the coupling law, or the map from reconstructed Krylov levels to physical HSMT grades.

Proposition 9.6 (Current-core experimental-target selection no-go [NG])**.** *Under the currently specified physical interpretations, the existing HSMT operators and projectors do not select a unique physically measurable probe/contact realization. A fundamental experimental target therefore requires independent physical input.*

9.6 Six-gap ledger

The missing physical structures can be organized into six conceptual categories:

M1 Hierarchy origin [C/O]: what determines why the hierarchy exists? C15 is one candidate.

M2 Microscopic Jacobi dynamics [P/O]: what determines the infinite coefficient law? Finite coefficients are reconstructible; the infinite law is not selected.

M3 Reference-to-physical geometry bridge [C]: why should C17a describe physical geometry? C17b is the candidate bridge.

M4 Full spatial soldering [O]: how are radial and tangential/support sectors combined into full space?

M5 R**-odd shape physics** [O]: what physical ingredient can distinguish shape endpoints inaccessible to the frozen R-even sector?

M6a Lorentzian/causal completion [O]: what produces an indefinite spacetime principal structure and temporal/spatial calibration?

M6b Temporal orientation/arrow [O]: what supplies a physical time orientation or irreversible arrow?

These gaps are irreducible relative to the currently frozen architecture in the limited sense that the present assumptions do not derive one from the others. A deeper future theory could still unify them through genuinely new cross-sector predictions.

Question	Status	Current result	Input needed to reopen
Unique asymptotic Jacobi law?	[NG]	Not selected by current core	Microscopic dynamics or asymptotic data
Centered resonance derived?	[NG]	Testable but not selected	Independent chiral-type principle or evidence
Unique ladder algebra?	[NG]	C22 remains a candidate class	Microscopic statistics/algebra principle or discrimination
Finite data fix asymptotic C17a geometry?	[NG]	No	Independent tail information
Does C17a imply physical space?	[C/O]	No bridge derived	Operational/physical identification
Does radial geometry fix full spatial geometry?	[NG]	No	Angular/support sector and soldering data
Lorentzian/time structure selected?	[NG]	No	Independent temporal/signature physics
Fundamental probe/contact target selected?	[NG]	No, relative to current core	Physical $P/Q/K$ identification

Table 1: Selection and identifiability boundaries under the current HSMT architecture.

10 Boundary Jacobi Spectroscopy

The inverse-spectral framework becomes operational if a measurable probe couples linearly to a cyclic hidden sector. This section derives the conditional response formula. It does not identify a fundamental physical HSMT probe.

10.1 Single-contact probe model

Let $|v\rangle$ be a normalized visible probe state with bare energy or frequency ω_0. Let J describe a hidden self-adjoint sector with normalized contact state $|e_0\rangle$. Assume the single-contact coupling

$$H_{\mathrm{tot}} = \omega_0 |v\rangle\langle v| + J + g_p\big(|v\rangle\langle e_0| + |e_0\rangle\langle v|\big).$$

Define

$$G_v(z) = \langle v, (z - H_{\text{tot}})^{-1} v \rangle.$$

The Schur complement gives the standard one-channel Feshbach response [3]:

$$G_v(z) = \frac{1}{z - \omega_0 - g_p^2 m_0(z)}. \tag{25}$$

Proposition 10.1 (Probe-response/boundary-m inversion [D])**.** *Under the linear single-contact model,*

$$m_0(z) = \frac{z - \omega_0 - G_v(z)^{-1}}{g_p^2}. \tag{26}$$

Thus a calibrated complex probe response determines the contact m-function.

10.2 Calibrated probe background

If the visible probe has a known intrinsic self-energy $\Sigma_{\text{probe}}(z)$, then

$$G_v(z)^{-1} = z - \omega_0 - \Sigma_{\text{probe}}(z) - g_p^2 m_0(z),$$

so

$$m_0(z) = \frac{z - \omega_0 - \Sigma_{\text{probe}}(z) - G_v(z)^{-1}}{g_p^2}.$$

Proposition 10.2 (Unknown probe background/hidden self-energy no-go [NG])**.** *Without independent probe-background calibration or additional analytic assumptions, a frequency-dependent visible-probe self-energy cannot generally be separated uniquely from the hidden-sector self-energy using the scalar response alone.*

10.3 Complex response requirement

The inversion uses the full complex G_v^{-1}. Magnitude-only response $|G_v|$ does not generically determine the analytic function G_v. Minimum-phase assumptions, global causality reconstruction, or sufficiently complete bandwidth information may restore phase under additional conditions.

Proposition 10.3 (Magnitude-only/m-function identification no-go [NG])**.** *Magnitude-only spectral response does not generically determine the boundary m-function uniquely.*

10.4 Tunable-coupling collapse

If the same hidden J can be measured at two calibrated couplings g_1, g_2 without otherwise changing the hidden sector, then

$$m_0(z) = \frac{G_{g_1}(z)^{-1} - G_{g_2}(z)^{-1}}{g_2^2 - g_1^2}. \tag{27}$$

More generally, the extracted hidden self-energy divided by g_p^2 should collapse onto the same $m_0(z)$ throughout the linear single-contact regime. Failure may indicate nonlinear response, probe-induced modification of J, multiple contact channels, calibration error, or breakdown of the assumed model.

10.5 Internal consistency checks

Any extracted contact function should satisfy, under the convention used throughout this paper,

$$\operatorname{Im} z > 0 \Longrightarrow \operatorname{Im} m_0(z) < 0, \qquad z\, m_0(z) \to 1,$$

and the reconstructed measure must be nonnegative and normalized. Coupling-collapse consistency and held-out response reconstruction provide additional diagnostics.

The exact formalism allows absolutely continuous, discrete, singular-continuous, or mixed measures. The numerical validation in A1 covers absolutely continuous benchmark densities and discrete atoms, not generic singular-continuous reconstruction.

10.6 Finite bandwidth and stable depth

Any actual response measurement covers finite bandwidth and finite resolution. Operationally, the supported output should be a finite prefix $\{b_N, \lambda_N\}_{N \le K_{\mathrm{stable}}}$ with covariance or error estimates.

Proposition 10.4 (Finite-bandwidth/infinite-Jacobi reconstruction no-go [NG])**.** *Finite-bandwidth, finite-resolution response data do not justify unique reconstruction of an infinite Jacobi chain without additional continuation assumptions. Operational reports should stop at a demonstrated stable depth.*

10.7 E1a: Boundary Jacobi Spectroscopy

[DC] **E1a, Boundary Jacobi Spectroscopy**, is the mature conditional spectroscopy framework consisting of:

1. identifying and calibrating a visible probe mode;
2. measuring its complex response;
3. removing known probe background;
4. extracting the contact m-function;
5. verifying analytic-sign and normalization consistency;
6. reconstructing the mixed contact spectral measure;
7. recovering a stable finite Jacobi prefix with covariance;
8. applying candidate-specific tests such as centered resonance and C22 finite-difference relations.

E1a is not itself a fundamental HSMT prediction.

10.8 Krylov chain versus HSMT grade chain

Scalar contact data determine the Jacobi/Krylov representation generated from the chosen contact state. They do not identify a pre-existing microscopic grade basis.

Proposition 10.5 (Contact-Krylov/physical-grade identification no-go [NG])**.** *Boundary spectral data determine the contact Krylov/Jacobi equivalence class, not an independently existing HSMT grade decomposition. An additional physical map is required before the recurrence index can be called a measured HSMT grade.*

We therefore distinguish:

E1a Boundary Jacobi Spectroscopy [DC]: calibrated response → contact measure → finite Jacobi/Krylov coefficients.

E1b Grade-Resolved Jacobi Spectroscopy [C/O]: additionally requires an independently justified Krylov-level/HSMT-grade identification.

E1-H Fundamental HSMT physical target [O]: additionally requires a physically identified probe, physical hidden sector, explicit coupling operator, calibrated contact law, grade/contact bridge, and pre-fit prediction.

Under the currently specified physical interpretations, the abstract operators D_ρ, D_{sp}, H_{eff}, P_N, K_N, the four-site shape sector, and the memory kernel do not identify a unique physically preparable and readable probe subsystem.

Proposition 10.6 (Current-core probe-selection no-go [NG])**.** *The presently frozen HSMT operators do not uniquely determine a physically measurable E1-H probe/contact system. The downstream Feshbach response machinery is ready once a physical $P/Q/K$ realization is supplied, but the current architecture does not supply that realization.*

An engineered oscillator or resonator network with a known hidden Jacobi operator can test calibration, response inversion, finite-prefix recovery, pole reconstruction, covariance propagation, and A2 candidate tests. Such an engineered analogue validates the spectroscopy method, not the claim that nature realizes a fundamental HSMT grade structure.

Proposition 10.7 (Analog validation/fundamental-HSMT validation no-go [NG])**.** *Successful Boundary Jacobi Spectroscopy on an engineered Jacobi network does not constitute evidence for a fundamental HSMT grade structure unless an independent physical target map is supplied.*

11 Discussion

The strongest positive results of the present analysis concern cyclic inverse spectral reconstruction, finite-prefix recovery, candidate-specific falsification, exact centered-symmetry relations, and conditional operator-induced reference geometry. The strongest negative results concern microscopic law selection, asymptotic-tail inference from finite data, physical-geometric interpretation, Lorentzian/temporal completion, and direct experimental-target selection.

The resulting framework is therefore neither a closed fundamental theory nor merely a collection of unconstrained hypotheses. It is a reconstruction-and-identifiability framework with explicit boundaries on what present assumptions can support.

11.1 Reconstruction, selection, interpretation

The central methodological distinction is

$$\text{reconstruction} \neq \text{selection} \neq \text{interpretation}.$$

For the cyclic Jacobi sector, reconstruction is mathematically controlled. Selection of a forward microscopic law is not. Physical interpretation is more underdetermined still. This hierarchy prevents an inverse reconstruction result from being promoted into a microscopic or geometric prediction without an additional bridge.

11.2 What finite spectral data provide

Under the assumptions of section 10, a suitable complex boundary response can be converted into a contact m-function. That function supports reconstruction of the contact spectral measure, stable finite Jacobi coefficients, diagonal detuning, candidate cross-level relations, and discrete pole data under appropriate conditions. The output is finite operator information, not an asymptotic theory.

11.3 Why no-go results are constructive

A no-go under current assumptions identifies where new information must enter. The asymptotic-tail selection no-go means that a future tail law must come from new microscopic dynamics, a new symmetry, or independently justified data rather than another rearrangement of the same assumptions. The C17a/physical-geometry no-go means that physical geometry requires an independent bridge rather than a reinterpretation of mathematical canonicity. Explicit underdetermination is therefore useful theoretical information.

11.4 The unresolved physical sectors

The M1–M6 ledger in section 9 organizes the remaining physical gaps. These categories are irreducible only relative to the currently frozen architecture. A future unification would have to generate nontrivial cross-sector predictions from fewer independent inputs rather than merely packaging multiple assumptions into one formal object.

M1 and M2 are the most natural potential compression point: a finitely presented ladder system could generate both hierarchy and cross-level dynamics. But finite generativity does not select one algebra. M3 and M4 remain distinct because even a mathematically coherent radial reference process does not supply the bridge to physical geometry or the angular/support soldering needed for full space. M5 remains open because the frozen R-even shape sector does not supply the missing R-odd selector. M6 remains open because a self-adjoint generator does not thereby produce Lorentzian spacetime structure or a temporal arrow.

11.5 Experimental status

Boundary Jacobi Spectroscopy is mathematically operational at the response-inversion level under its stated assumptions, but the present theory does not yet identify a fundamental physical HSMT target. The current architecture does not provide a unique physical P-sector, Q-sector, coupling K, or Krylov-to-grade map. An engineered analogue can therefore test the spectroscopy framework while leaving the fundamental HSMT target question open.

11.6 What this paper does not claim

This work does not derive a physical spatial dimension, physical three-space, a $3+1$ spacetime, Lorentzian signature, a temporal arrow, a unique microscopic ladder algebra, a unique asymptotic Jacobi law, a fundamental resonance principle, binary shape selection, a physical interpretation of C17a, a value of α_{H}, a cosmological redshift mechanism, a CMB shell boundary, a dark-matter or dark-energy explanation, or a fundamental experimental HSMT target. Historical derivational chains not re-established under the present status rules are not used as results here.

11.7 What would count as progress beyond this paper

Future progress should enter through genuinely new physical input or genuinely new empirical information. Examples of the former include an independently motivated microscopic ladder algebra, a physical geometry bridge, an R-odd shape interaction, a Lorentzian/clock structure, or an experimentally identified $P/Q/K$ coupling. Examples of the latter include a calibrated complex response yielding a stable finite Jacobi prefix, resolved diagonal detuning, rejection of a candidate ladder relation, or an independently established contact/grade map. Recombining assumptions already shown to admit explicit counterfamilies does not by itself advance the theory.

12 Conclusion

This paper isolated the reconstruction and identifiability content of the HSMT operator architecture.

First, for a normalized cyclic contact sector represented by a self-adjoint Jacobi operator, the exact boundary m-function determines the corresponding contact spectral measure and Jacobi recurrence coefficients. Under finite-resolution and noisy conditions, the practically supported output is a stable finite Jacobi prefix rather than an unrestricted infinite continuation.

Second, the reconstructed coefficients support direct falsification tests for candidate structure. Centered spectral symmetry is equivalent, within the cyclic Jacobi representation, to a constant diagonal about one energy center. Finite-generativity ladder models such as Toeplitz, CCR, and $\mathrm{SU}(1,1)$ generate distinct cross-level relations, although finite-prefix data do not generally identify a unique microscopic algebra.

Third, several stronger conclusions do not follow from the present assumptions. The current architecture does not select an asymptotic Jacobi law, centered resonance, a microscopic ladder algebra, physical geometry from C17a, a full spatial metric, Lorentzian/temporal structure, or a fundamental experimental probe/contact system. These are not filled in by preferred continuation; they are recorded as explicit open or no-go boundaries.

The scientific posture is therefore deliberately asymmetric: finite operator reconstruction is mature within the stated mathematical and benchmark conditions, while microscopic and physical interpretation remains incomplete. Further progress requires either genuinely new physical principles or new empirical information. The present framework supplies the operator and spectral machinery needed to test such future input without confusing reconstruction with selection or interpretation.

Summary of logical status

Result or structure	Status	Established	Not established
Exact cyclic inverse reconstruction	[D]	Exact contact m-function/measure determines the associated Jacobi coefficients	No forward microscopic law selected
Application A1	[R/D/NG]	Finite-prefix recovery in declared benchmarks and explicit identifiability limits	No infinite-tail determination
Centered spectral symmetry	[D]	Equivalent to constant cyclic Jacobi diagonal about one center	No fundamental resonance principle
Centered Resonant Immanence	[C]	Falsifiable candidate	Not selected by current HSMT core
C22 finite-generativity class	[C]	Search class with falsifiable cross-level predictions	No unique algebra selected
Application A2	[D/DC/R/NG]	Candidate-specific finite-prefix falsification	No unique microscopic identification
C17a	[DC]	Conditional operator-induced reversible radial reference construction	No physical-space interpretation
C17b	[C]	Physical-geometry bridge candidate	Not derived
Boundary Jacobi Spectroscopy	[DC]	Conditional complex-response-to-Jacobi reconstruction	No fundamental HSMT target identified
E1-H	[O]	Experimental-interface problem formulated	No physical target presently selected
Six-gap ledger	[NG]	Relative to the current architecture, the missing sectors do not collapse	Future deeper unification not excluded

Reproducibility statement

All numerical claims in Applications A1 and A2 are synthetic benchmark results. The manuscript reports the frozen parameter families, key resolutions and noise levels, held-out errors, pole data, and representative discrimination behavior required to interpret those claims. A submission-quality reproducibility package should contain the exact sample grids, optimization settings, terminator variants, covariance calculations, random seeds or seed policy, full Monte Carlo tables, and code used to generate the reported figures and statistics. No benchmark parameter is interpreted as a physical HSMT prediction.

Data and code availability

This manuscript is assembled from a frozen theoretical and synthetic-benchmark record. No external experimental dataset is analyzed. Code and generated benchmark data should accompany the submission in a versioned repository or archival supplement once the final implementation bundle is deposited.

References

[1] G. Teschl, *Jacobi Operators and Completely Integrable Nonlinear Lattices*, Mathematical Surveys and Monographs, Vol. 72, American Mathematical Society, Providence, 2000.

[2] T. S. Chihara, *An Introduction to Orthogonal Polynomials*, Gordon and Breach, New York, 1978.

[3] H. Feshbach, "Unified theory of nuclear reactions," *Annals of Physics* **5** (1958), 357–390, doi:10.1016/0003-4916(58)90007-1.

A Jacobi moments and recurrence identities

For completeness, with normalized e_0,

$$\mu_1 = b_0, \qquad \mu_2 = b_0^2 + \lambda_0^2,$$

$$\mu_3 = b_0^3 + 2b_0\lambda_0^2 + b_1\lambda_0^2.$$

Let $\beta_0 = \mu_1 = b_0$ and

$$\nu_n = \int (E - \beta_0)^n \, d\mu(E).$$

Then

$$\nu_2 = \lambda_0^2, \qquad \nu_3 = \lambda_0^2(b_1 - b_0),$$

and hence, for $\nu_2 > 0$,

$$b_1 - b_0 = \frac{\nu_3}{\nu_2}.$$

These identities are invariant under a global shift $J \mapsto J + cI$ once expressed through diagonal differences.

B A1 numerical protocol

The frozen A1 record uses complex boundary-resolvent data evaluated at finite imaginary resolution. The principal declared resolution is $\eta = 0.02$, with additional tail-comparison calculations at $\eta = 0.005$ and 0.05. Synthetic complex response noise is tested at fractional scales 10^{-4}, 10^{-3}, and 10^{-2}.

For finite-rank benchmarks, the forward map is the exact continued fraction built from equation (8). For long-tail tests, a deep backward continued fraction generates the synthetic truth. Candidate finite prefixes are fit and evaluated against held-out complex-response values. The stable depth

K_{stable} is defined operationally by contiguous coefficient stability under noise perturbation, model-depth variation, and held-out validation. The specific threshold is a benchmark convention, not a theorem.

The frozen project record retains the principal numerical outcomes used in the main text: $K = 6$ and $K = 10$ held-out relative m-errors for the long-tail truth, the explicit Tail A/Tail C differences, and the A1-P pole/residue noise statistics. Exact implementation grids and optimizer settings belong in the companion reproducibility bundle and should be versioned with any submitted manuscript.

C A1 long-tail nonuniqueness construction

Tail A is

$$\lambda_N^A = 1 - \frac{0.35}{(N+2)^{1.3}}, \qquad b_N^A = \frac{0.15}{(N+2)^{1.1}}.$$

Tail C is identical for $N = 0, \ldots, 9$ and then continues as

$$\lambda_N^C = 1 - \frac{0.0542852834}{(N+2)^{0.55}}, \qquad b_N^C = \frac{0.0555160759}{(N+2)^{0.70}}.$$

The two asymptotic families are inequivalent despite an identical ten-layer prefix. Their declared finite-resolution boundary-response differences are

η	relative m-difference
0.005	5.08×10^{-4}
0.02	3.89×10^{-4}
0.05	2.44×10^{-4}

and lie below the tested 10^{-3} and 10^{-2} noise scales.

D A1-P off-support pole calculation

For

$$\lambda_\infty = 1, \quad \lambda_0 = 0.8, \quad \lambda_1 = 1.15, \quad b_0 = 2.5, \quad b_1 = -0.1,$$

the free essential band is $[-2, 2]$. The exact boundary function is

$$m_0(z) = \frac{1}{z - b_0 - \lambda_0^2 \left[z - b_1 - \lambda_1^2 m_\infty(z)\right]^{-1}}.$$

An off-band atom occurs at a real zero of the denominator outside $[-2, 2]$. The frozen calculation gives

$$E_p \approx 2.77656525146, \qquad w_p \approx 0.86623881586.$$

With the convention equation (2), the atomic weight is the residue

$$w_p = \operatorname*{Res}_{z=E_p} m_0(z),$$

which is positive. The remaining continuous mass is approximately 0.13376118400. Finite η replaces the point mass by a Lorentzian-broadened contribution, so exact pole identification must not be conflated with peak detection at one finite resolution.

E A2 Monte Carlo discrimination

The coefficient-level synthetic observation model is

$$\lambda_N^{\mathrm{obs}} = \lambda_N(1+\epsilon_N), \qquad \epsilon_N \sim \mathcal{N}(0,\sigma_\lambda^2),$$

with representative $\sigma_\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}\}$. The fitted quantities are $x_N = (\lambda_N^{\mathrm{obs}})^2$. The benchmark compares minimal Toeplitz, CCR, and normalized SU(1,1) candidate laws over finite prefixes.

The frozen record establishes the following representative behavior: minimal Toeplitz is selected essentially perfectly in the tested cases through one-percent coefficient noise; a strongly curved SU(1,1) example such as $k = 1$ is easily separated; for $k = 50$ at one-percent coefficient noise the SU(1,1) selection rate rises strongly with depth, from roughly 37% at a short prefix to about 99.8% by $K = 10$ in the declared scoring protocol; and for $k = 200$ at the same noise level the problem remains difficult, with only roughly 20–34% SU(1,1) selection through the tested depths. These frequencies are protocol-dependent benchmark statistics, not physical probabilities.

For covariance-aware curvature testing, use $c = D_2 x$ and $V = D_2 \Sigma_x D_2^T$. When V is nonsingular and well conditioned, use equation (15); otherwise use a pseudoinverse or regularized generalized least-squares procedure. If positive curvature is unresolved, report an uncertainty interval or a lower bound on k rather than declaring exact CCR.

F Specialized C17a asymptotics

Consider the narrower asymptotic family

$$\frac{\lambda_N}{\lambda_\infty} = 1 - \frac{c}{N^2} + o(N^{-2}), \qquad b_N = 0,$$

and seek a formal positive edge solution at $E_* = 2\lambda_\infty$ of the form

$$h_N \sim N^a.$$

Expanding the recurrence to leading order gives the indicial relation

$$a(a-1) = 2c,$$

with formal branches

$$a_\pm = \frac{1 \pm \sqrt{1+8c}}{2}.$$

This is a conditional asymptotic relation inside the stated family. The coefficient c alone does not determine which branch is selected by the full half-line boundary problem. The global positive edge solution depends on the complete operator and boundary condition.

G Additional no-go counterfamilies

This appendix collects compact witnesses used to delimit current-core selection claims.

Asymptotic Jacobi tails

Bounded positive examples compatible with the current structural assumptions include

$$\lambda_N = 1, \qquad \lambda_N = 1 + \frac{0.2}{N+2}, \qquad \lambda_N = 1 + 0.2(-1)^N,$$

$$\lambda_N = 1 + 0.2\,\mathrm{sinlog}(N+2), \qquad \lambda_N = 1 + \frac{0.2}{(N+2)^{1/2}}.$$

Their incompatible asymptotics demonstrate that the current structural assumptions do not select one tail law.

Centered resonance

For any admissible hopping sequence and bounded nonconstant real diagonal b_N,

$$J_b = A_\lambda + A_\lambda^* + \sum_N b_N P_N$$

preserves the frozen grade/Krylov organization and real self-adjoint structure while violating centered chiral symmetry.

Ladder normalization

The same basis can support

$$S|N\rangle = |N+1\rangle, \qquad a^\dagger|N\rangle = \sqrt{N+1}|N+1\rangle, \qquad K_+|N\rangle = \sqrt{(N+1)(N+2k)}|N+1\rangle.$$

Thus a hierarchy alone does not select a successor normalization.

Temporal completion

A positive operator L can consistently enter inequivalent evolutions such as

$$i\partial_t\psi = L\psi, \qquad (-\partial_t^2 + c^2 L)\phi = 0, \qquad \partial_\tau u = -Lu.$$

The common spatial/reference operator therefore does not select Lorentzian signature, a causal cone, or a temporal arrow.